Ecological Evaluation of Sunder Nursery

A field-based valuation study using CVM

RDL725 - Ecological Perspective of Growth and Development

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Roadmap

1. Why Sunder Nursery matters
2. Site background and ecological context
3. Method: Contingent Valuation Method (CVM)
4. Survey design and sample profile
5. CVM results: WTP, non-use values, ecological perception
6. Total Economic Value estimate
7. Limitations, conclusion, and field-visit proof

Why Did We Choose Sunder Nursery?

Why this site matters:

- Urban parks create recreational, ecological, cultural, and aesthetic benefits
- Most of these benefits are not directly priced in markets
- Therefore, their contribution is often undervalued in policy and management decisions

What we wanted to estimate:

- Estimate the value people attach to conserving Sunder Nursery
- Use **Contingent Valuation Method (CVM)** – it captures both use and non-use values through stated preferences
- CVM is well-suited because conservation benefits are not reflected in market prices

Fieldwork Snapshot

- Site: Sunder Nursery, New Delhi
- Visit date: 10 April 2026
- Time: 4:00 PM - 7:30 PM
- Respondents: 45
- Mode: in-person bilingual survey

Sunder Nursery: Site Background

Historical and ecological importance:

- 90-acre heritage park in the Nizamuddin area, next to Humayun's Tomb
- Originated as a Mughal-era nursery
- Restored by the Aga Khan Trust for Culture and reopened in 2018
- Includes restored monuments, wetlands, tree cover, flowering beds, and bird habitat

Why CVM is suited here (textbook criteria):

- **Uniqueness**: Mughal-era heritage + restored ecology – one of a kind
- **Irreversibility**: if lost, monuments and habitat cannot be recreated
- **Non-use value**: has both direct users and intergenerational significance

Garden of Delight at dusk

Flowering beds at Sunder Nursery

How We Evaluated the Site: CVM

What is CVM?

- A survey-based technique that creates a hypothetical market to elicit willingness to pay (WTP)
- Respondents are asked whether they would pay a higher fee for conservation
- This directly captures stated WTP for protection of monuments, gardens, and biodiversity

Why CVM?

- CVM captures **both** use and non-use values (Krutilla 1967; Johansson 1990)
- Other methods (market pricing, hedonic, TCM) cannot capture non-use values like bequest and existence value

TEV Framework (Textbook)

Total Value = Use Value + Non-Use Value

Non-Use Value = Option Value + Bequest Value + Existence Value

CVM is the only standard technique that captures all of these.

Non-Use Values We Captured

- Bequest value (preservation for future generations)
- Existence value (support even without future visits)
- Ecological condition perception

Why CVM Over Other Methods?

	CVM (our choice)	TCM (alternative)
What it measures	Stated WTP via hypothetical market	Revealed preference via travel cost
Use values	✓ Yes	✓ Yes
Non-use values (bequest, existence)	✓ Yes – this is its key strength	× No – cannot capture
Suitability here	Strong – park has heritage, ecological, and intergenerational value	Weak – 53% visitors had multi-destination trips

Textbook (Ch. on CVM): "Other approaches stress exclusively on estimating use values... Contingent valuation [is a] general technique to elicit willingness to pay in broad and inclusive sense – use and non-use value."

Survey Design (Following Textbook Two-Phase Approach)

Phase 1 – Scenario presentation:

1. Aspects of the park visitors value
2. Ecological condition rating

Phase 2 – Valuation questions:

3. WTP for a slightly higher entry fee
4. Maximum extra amount willing to pay
5. Reason for refusing (protest bid identification)
6. Bequest value: future generations
7. Existence value: support without future visits
8. Demographics: age, visit frequency, income

Design choices:

- Closed-ended bids (Rs. 10, 20, 50, 100) – textbook: "more accurate with closed-end choices"
- Payment vehicle: higher entry fee – realistic and neutral
- Modelled after the Kalamazoo Nature Center case study in the textbook

Survey Design Checks

1. Demographics and socioeconomic data collected
2. Protest-bid identification included
3. Relevant population: on-site visitors, n=45
4. Mode: in-person interviews

Data Collected

- Sample size: 45
- Date: Friday, 10 April 2026
- Time: 4:00 PM – 7:30 PM
- In-person bilingual mobile form

Our Survey Form

Scan to view the actual Google Form

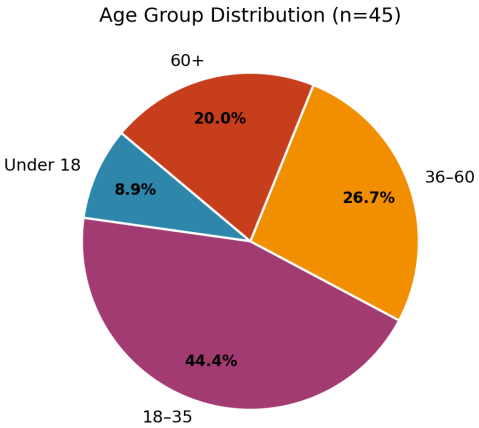
This is the bilingual survey form we used:

- Created as a Google Form in Hindi and English
- Administered in-person on mobile phones
- Visitors filled it out themselves after a brief explanation
- 16 questions covering valuation, travel, and demographics

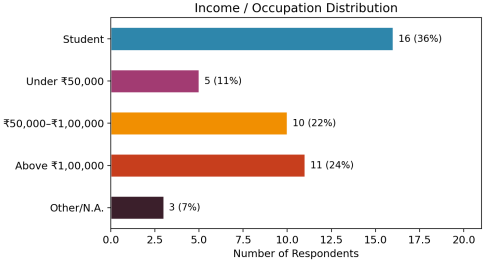
Response Summary

- Total responses collected: **45**
- All collected on-site at Sunder Nursery
- Date: 10 April 2026, 4:00–7:30 PM

Who We Surveyed



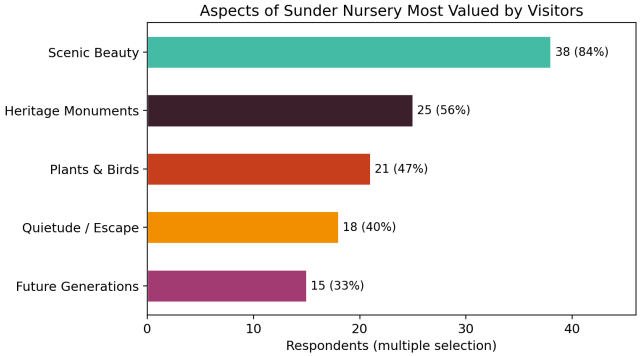
Age group distribution



Income/occupation distribution

Observation: Most respondents were young adults, and students formed the largest single group.

What Visitors Said They Value Most



Most-valued aspects of Sunder Nursery

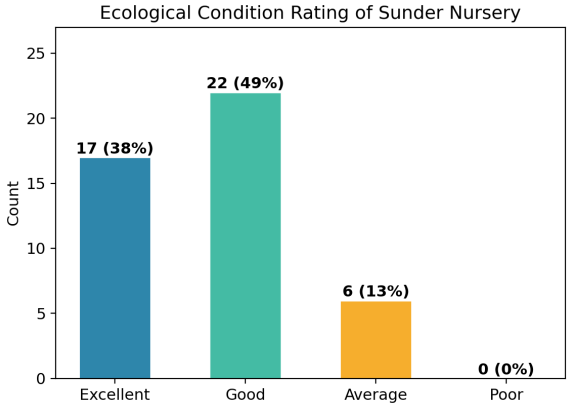
Top responses:

- Scenic beauty: 84.4%
- Heritage monuments: 55.6%
- Plants and birds: 46.7%
- Quietude / escape: 40.0%
- Future generations: 33.3%

Our reading of this result

People were not valuing only the monuments or only the greenery. What stands out is the combined heritage-plus-ecology experience.

How Visitors Rated the Ecological Condition



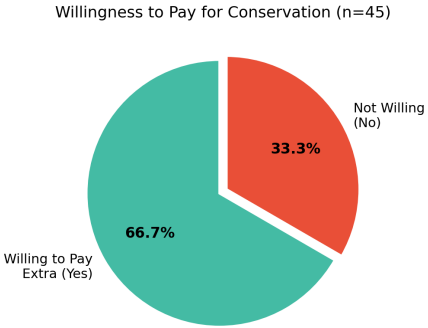
Ecological condition rating

Responses:

- Excellent: 37.8%
- Good: 48.9%
- Average: 13.3%
- Poor: 0.0%

A combined 86.7% rated the nursery as Good or Excellent, which supports the view that the restoration is visible to visitors.

CVM Result 1: Willingness to Pay



Would visitors pay a higher entry fee?

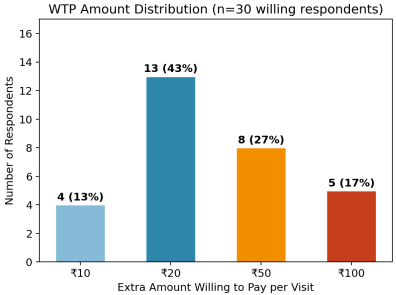
Measure	Value
Willing to pay more	30 / 45 (66.7%)
Not willing	15 / 45 (33.3%)
Mean WTP (willing)	Rs. 90
Median WTP (willing)	Rs. 70
Overall mean WTP	Rs. 76.7

WTP = Current fee + Extra amount

Current entry fee = Rs. 50. Survey asked: "*maximum extra amount*". By definition, WTP = total amount a person is willing to pay.

Willing: WTP = Rs. 50 + mean extra (Rs. 40) = **Rs. 90**
Unwilling: WTP = Rs. 50 (they are visiting, so WTP $\geq$ Rs. 50)
Overall: $\frac{(30\times 90)+(15\times 50)}{45}$ = **Rs. 76.7**

CVM Result 2: WTP Distribution and Protest Bids



Extra amount above Rs. 50 ticket → Total WTP:

Extra	Total WTP	n
+Rs. 10	Rs. 60	4
+Rs. 20	Rs. 70	13
+Rs. 50	Rs. 100	8
+Rs. 100	Rs. 150	5

Among the 15 "No" responses:

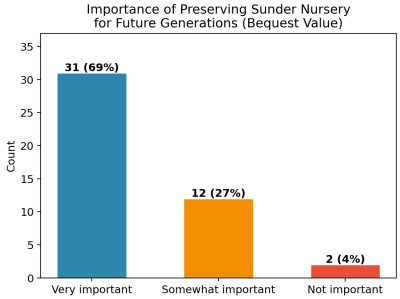
- 40% (6): not worth paying more
- 33% (5): government should pay
- 27% (4): cannot afford

Protest-bid adjustment

5 "government should pay" = protest bids (they value the park but reject the payment vehicle). Adjusted: 30/40 = 75% WTP rate.

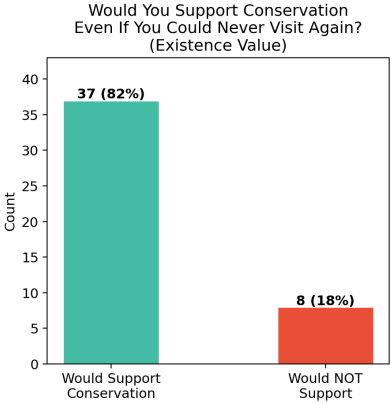
Rs. 70 (median) is the most representative WTP – the majority chose modest bids.

Non-Use Value Responses



Bequest value

68.9% said preservation is **very important** for future generations

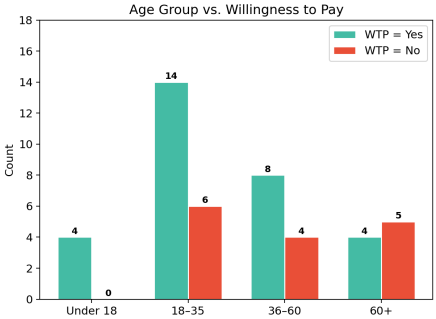


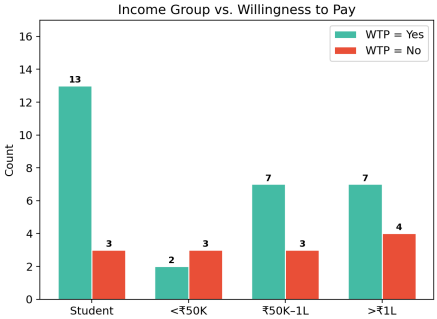
Existence value

82.2% would support conservation even if they could never visit again

Observation: These responses show that people value the nursery even beyond their own present-day use.

Cross-Tabulations: Who is Willing to Pay?





Age	WTP %	n
Under 18	100%	4/4
18–35	70%	14/20
36–60	67%	8/12
60+	44%	4/9

Income	WTP %	n
Student	81%	13/16
<50k	40%	2/5
50k–1L	70%	7/10
≥1L	62%	8/13

Younger visitors and students show the highest WTP rate despite lower incomes – suggesting conservation motivation is not income-driven.

Total Economic Value from CVM

TEV Framework (Textbook)

Total Value = Use Value + Non-Use Value

Non-Use = Option + Bequest + Existence

CVM captures all of these through stated WTP (Schulze et al. 1981; Walsh et al. 1984).

CVM Estimate	Value
Overall mean WTP	Rs. 76.7/visit
Aggregate annual WTP	**Rs. 11.50 cr/yr**
Current ticket revenue	Rs. 7.50 cr/yr
Additional ecological value	**Rs. 4.00 cr/yr**
Bequest value (very important)	68.9%
Existence value (would support)	82.2%

Aggregate WTP:
$\text{Rs. } 76.7 \times 15{,}00{,}000 = \textbf{Rs. 11.50}$ **crore/year**

Value already captured by market:
$\text{Rs. } 50 \times 15{,}00{,}000 = \text{Rs. } 7.50$ crore (ticket revenue)

Ecological value above market price:
$\text{Rs. } 11.50 - \text{Rs. } 7.50 = \textbf{Rs. 4.00}$ **crore/year**

What this tells us:
The Rs. 4.00 crore is the value visitors place on Sunder Nursery's ecological and heritage conservation *above* what the current ticket price captures. This includes:

- **Use value** – enjoyment of gardens, monuments, greenery
- **Non-use value** – bequest (68.9%) and existence (82.2%)

Conservative estimate: bids capped at Rs. 100 extra, and only on-site visitors were surveyed.

Discussion and Limitations

What we can say from the CVM data:

- Visitors value the site well beyond the entry fee
- Ecological condition is viewed positively (86.7% Good/Excellent)
- A majority (66.7%) is willing to pay more for conservation
- Strong non-use values: bequest (68.9%) and existence (82.2%)

What this suggests:

- A modest fee increase appears acceptable to most visitors
- Conservation spending is justified from both visitor preference and ecological perspectives

CVM Limitations (from textbook)

- **Hypothetical bias:** WTP is stated, not actually paid
- **Strategic bias:** some may under/over-state (free-rider problem)
- **Information bias:** scenario description may influence responses
- Sample: 45 respondents, single Friday evening
- Closed-ended bids cap at Rs. 100 extra

Despite these, the results are internally consistent: WTP, non-use values, and ecological perception all align.

Conclusion

Main message

Using CVM, we estimate a total WTP of Rs. 11.50 crore/year (Rs. 76.7/visitor). Since the current ticket captures only Rs. 7.50 crore, the remaining Rs. 4.00 crore/year represents ecological and heritage value not reflected in the market price.

1. 66.7% willing to pay more; mean total WTP = Rs. 90 (willing), Rs. 76.7 (overall)
2. Strong non-use values: 68.9% bequest, 82.2% existence
3. 86.7% rated ecological condition as Good or Excellent
4. Rs. 4.00 crore/year in ecological value above market price

Final takeaway

Sunder Nursery's true value is significantly higher than what its entry fee captures. CVM reveals this hidden ecological and heritage value systematically.

References

[1] Krutilla, J.V. (1967). Conservation reconsidered. *American Economic Review*, 57(4), 777–786.

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[3] Walsh, R.G., Loomis, J.B., and Gillman, R.A. (1984). Valuing option, existence, and bequest demands for wilderness. *Land Economics*, 60(1), 14–29.

[4] Schulze, W.D., d'Arge, R.C., and Brookshire, D.S. (1981). Valuing environmental commodities: Some recent experiments. *Land Economics*, 57(2), 151–172.

[5] Dixon, J.A. and Hufschmidt, M.M. (1986). *Economic Valuation Techniques for the Environment*. Johns Hopkins University Press.

[6] Aga Khan Trust for Culture (2018). *Sunder Nursery Heritage Park*.

[7] RDL725 Course Material, IIT Delhi.

Field Visit Proof and Response Collection

Interacting with visitors

Conducting the survey on-site

With respondents at the amphitheatre

Surveying near the garden

Thank You

We visited the site and collected all responses in person. Open for Questions.